$$\hat{H} = \frac{1}{2}\omega_0\hat{\sigma}_z + \sum_i \omega_i \hat{a}_i^\dagger \hat{a}_i + \hat{\sigma}_z \otimes \sum_i \left(g_i \hat{a}_i^\dagger + g_i^* \hat{a}_i \right), \tag{5.90}$$